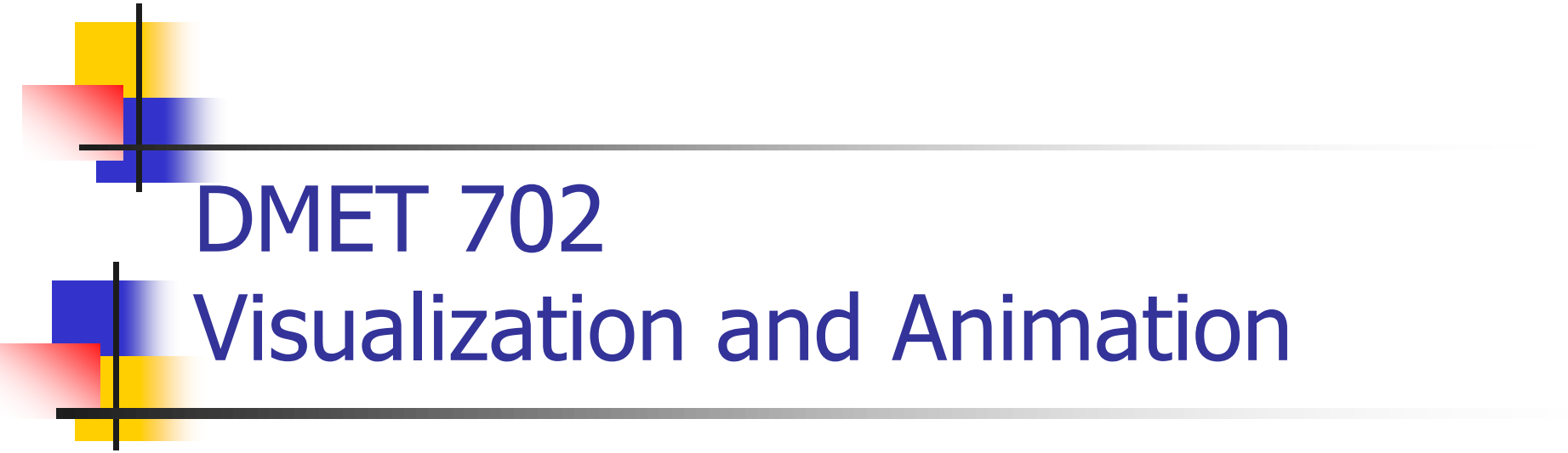


DMET 702
Visualization and Animation

Ray Casting

Assoc. Prof. Dr. Rimon Elias

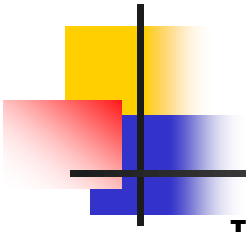
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  - Ray intersections
    - Ray-plane intersection
    - Ray-sphere intersection
    - Ray-triangle intersection
  - Accelerating the process
    - Bounding volume
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    - Quadtrees
    - Octrees

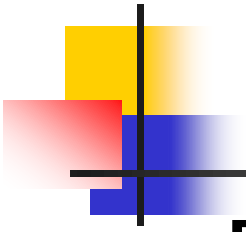


Sec 10.6

Ray Casting vs. Ray Tracing



- In order to find out which space point contributes to which pixel in the generated image, we may use ray casting and ray tracing.
- **Ray casting** and **ray tracing** are methods used to produce 2D images for 3D models.
- Both **ray casting** and **ray tracing** use rays that go from the center of projection (COP) to objects in 3D space passing through the image plane.
- However, differences do exist between them!

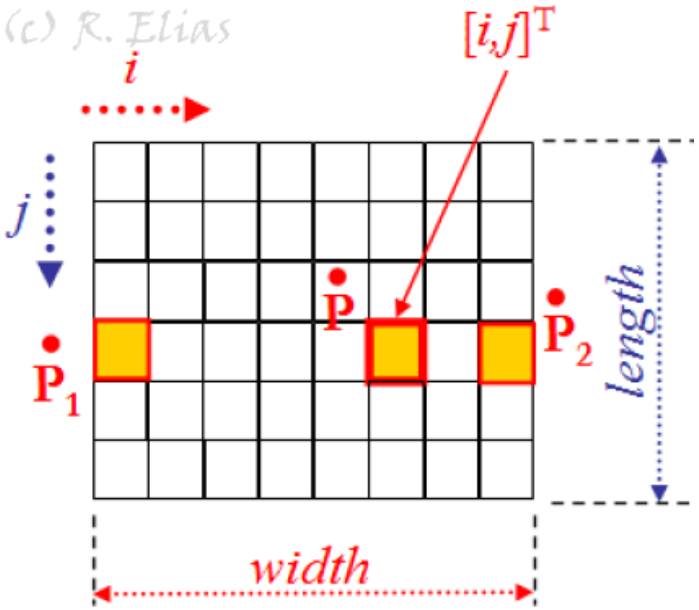


Ray Casting

- **Ray casting** is a technique used to determine the objects seen in an image through a camera by detecting objects intersected by rays.
  - You may think of the image as a window that you use to see what is in the scene.
  - Emit rays from the camera location (i.e., the center of projection) to intersect each and every pixel in the image.
  - Extend each ray until it hits an object in the scene.
  - The point of intersection is the location of what should be seen through that pixel.
  - The color assigned to the pixel under consideration depends on the radiance emanating from the surface of the object at the point of intersection.

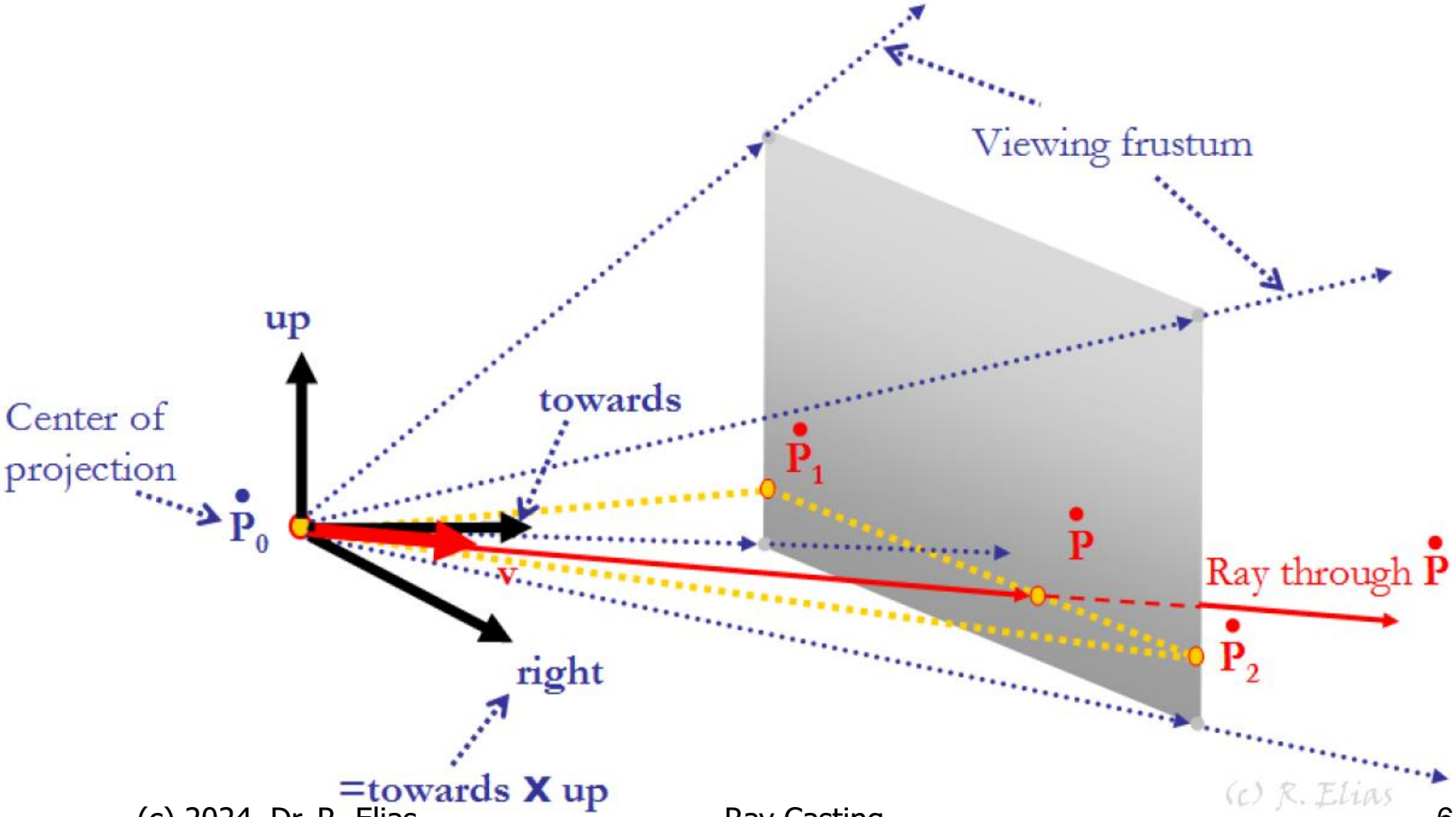
Rays through Pixels

- A ray is constructed for each pixel.
- In order to determine the equation for the ray, you need to specify
  - the location of the camera in space, which is known (i.e., $\dot{P}_0$);
  - the location of the pixel in space (i.e., $\dot{P}$), which can be estimated.



The observed image

Rays through Pixels

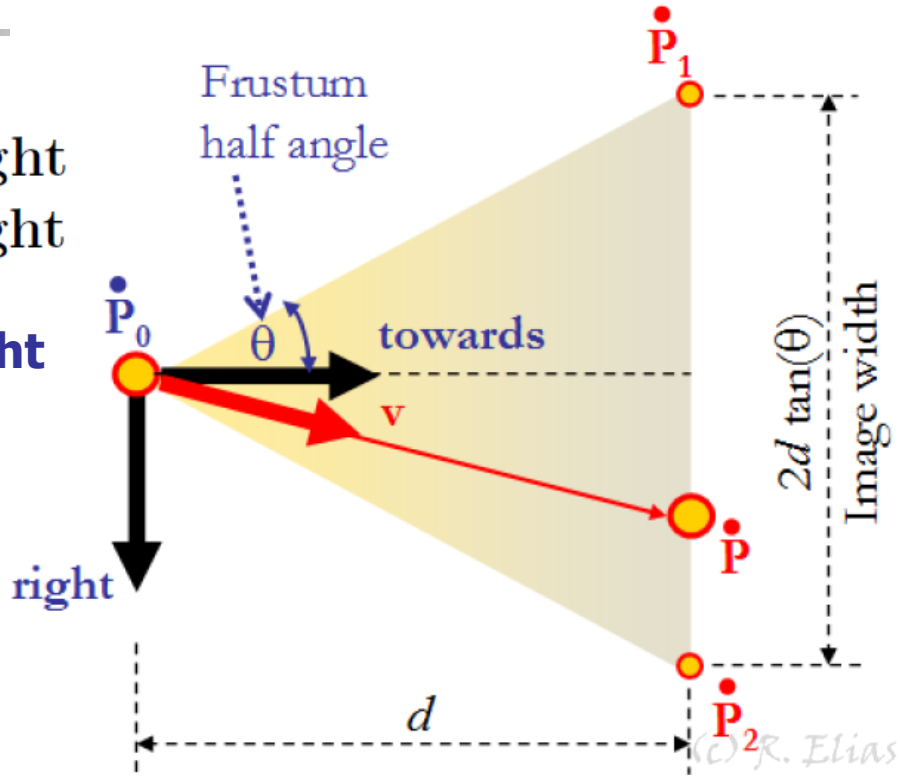


Rays through Pixels

$$\begin{aligned}\mathbf{P}_1 &= \mathbf{P}_0 + d \text{ towards} - d\tan(\theta) \text{ right}\\ \dot{\mathbf{P}}_2 &= \dot{\mathbf{P}}_0 + d \text{ towards} + d\tan(\theta) \text{ right}\end{aligned}$$

2d tan(θ) **right**

$$\dot{\mathbf{P}} = \dot{\mathbf{P}}_1 + \left(\frac{i+0.5}{width}\right)\left(\dot{\mathbf{P}}_2 - \dot{\mathbf{P}}_1\right)$$



Ray Equation

$$\begin{aligned}\dot{\mathbf{P}} &= \dot{\mathbf{P}}_0 + t\left(\frac{\dot{\mathbf{P}} - \dot{\mathbf{P}}_0}{\|\dot{\mathbf{P}} - \dot{\mathbf{P}}_0\|}\right)\\ &= \dot{\mathbf{P}}_0 + t\mathbf{v}\end{aligned}$$

What's the difference between 2d tan(θ) and width?

Ray Intersections: Examples of Objects

- Rays must be tested for intersections with objects in the scene.

- Examples of objects:
  - Triangles
  - Other polygons (concave or convex)
  - Boxes
  - Spheres
  - Cones
  - Cylinders
  - …

Ray-Plane Intersection

- A plane equation is expressed as

$$ax + by + cz = d,$$

where **n** = [*a, b, c*]^T is the normal vector to the plane.

- In case that **n** is normalized to a unit magnitude, *d* becomes the perpendicular distance from the origin to the plane.
- If the point of intersection is $\dot{\mathrm{P}}$, then

$$\dot{\mathrm{P}} \bullet \mathbf{n} = d \quad \text{or} \quad \left[\dot{\mathrm{P}}_0 + t\mathbf{v}\right] \bullet \mathbf{n} = d$$

- Thus,

$$t = \frac{-\left(\dot{\mathrm{P}}_0 \bullet \mathbf{n}\right) + d}{\mathbf{v} \bullet \mathbf{n}}$$

Intersection point

$$\dot{\mathrm{P}} = \dot{\mathrm{P}}_0 + t\mathbf{v}$$

Ray-Plane Intersection: An Example

- **Example:** Determine the point of intersection between a ray and a plane. The ray is going through two points; $\dot{\mathbf{P}}_0 = [3.0, 4.0, 5.0]^T$ and $\dot{\mathbf{P}}_1 = [5.0, -1.5, 4.0]^T$ and the plane equation is given by $6.0x - 2.0y + 1.5z - 4.0 = 0.0$. (Also, the plane equation can be expressed by the vector $[6.0, -2.0, 1.5, -4.0]^T$.)

- **Answer:** The direction vector **v** is given by

$$\begin{aligned}\mathbf{v} &= \dot{\mathbf{P}}_1 - \dot{\mathbf{P}}_0 \\ &= \begin{bmatrix} 5.0 \\ -1.5 \\ 4.0 \end{bmatrix} - \begin{bmatrix} 3.0 \\ 4.0 \\ 5.0 \end{bmatrix} = \begin{bmatrix} 2.0 \\ -5.5 \\ -1.0 \end{bmatrix}\end{aligned}$$

Ray-Plane Intersection: An Example

- The parameter t is obtained as

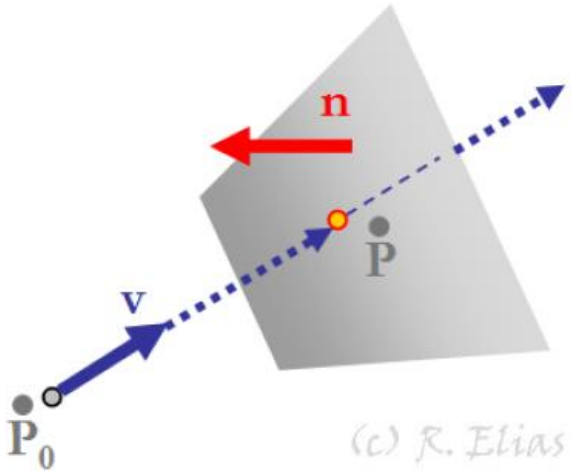
$$t = \frac{-(\dot{\mathbf{P}}_0 \bullet \mathbf{n}) + d}{\mathbf{v} \bullet \mathbf{n}} = \frac{-\left(\begin{bmatrix} 3.0 \\ 4.0 \\ 5.0 \end{bmatrix} \bullet \begin{bmatrix} 6.0 \\ -2.0 \\ 1.5 \end{bmatrix}\right) + 4.0}{\begin{bmatrix} 2.0 \\ -5.5 \\ -1.0 \end{bmatrix} \bullet \begin{bmatrix} 6.0 \\ -2.0 \\ 1.5 \end{bmatrix}} = -0.6279$$

- Then the point of intersection is estimated as

$$\begin{aligned} \dot{\mathbf{P}} &= \dot{\mathbf{P}}_0 + t\mathbf{v} \\ &= \begin{bmatrix} 3.0 \\ 4.0 \\ 5.0 \end{bmatrix} - 0.6279 \begin{bmatrix} 2.0 \\ -5.5 \\ -1.0 \end{bmatrix} = \begin{bmatrix} 1.7442 \\ 7.4535 \\ 5.6279 \end{bmatrix} \end{aligned}$$

Ray-Triangle Intersection

- In order to detect if a ray intersects a triangle
  1. Check for the intersection point with the plane of the triangle. (Refer to the previous slides.)
  2. If a point exists, check if its location falls inside the boundaries of the triangle.

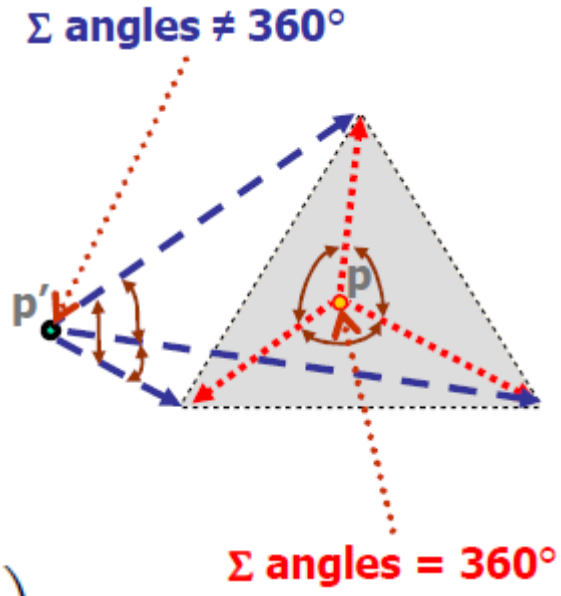


Ray-Triangle Intersection: Point in Triangle I

Using sum of angles and dot product

- To check if a point $\dot{P}$ falls inside a triangle:
  1. Construct a direction vector from point $\dot{P}$ to each of the vertices $\dot{P}_1, \dot{P}_2$ and $\dot{P}_3$. These vectors are $\dot{P}_1 - \dot{P}$, $\dot{P}_2 - \dot{P}$ and $\dot{P}_3 - \dot{P}$.
  2. Using the dot product, get the angles between the vectors. The angles are given by

$$\theta_3 = \cos^{-1}\left(\frac{[\dot{P}_1-\dot{P}]\bullet[\dot{P}_2-\dot{P}]}{\|\dot{P}_1-\dot{P}\|\|\dot{P}_2-\dot{P}\|}\right)$$
$$\theta_1 = \cos^{-1}\left(\frac{[\dot{P}_2-\dot{P}]\bullet[\dot{P}_3-\dot{P}]}{\|\dot{P}_2-\dot{P}\|\|\dot{P}_3-\dot{P}\|}\right)$$
$$\theta_2 = \cos^{-1}\left(\frac{[\dot{P}_3-\dot{P}]\bullet[\dot{P}_1-\dot{P}]}{\|\dot{P}_3-\dot{P}\|\|\dot{P}_1-\dot{P}\|}\right)$$



Ray-Triangle Intersection: Point in Triangle I

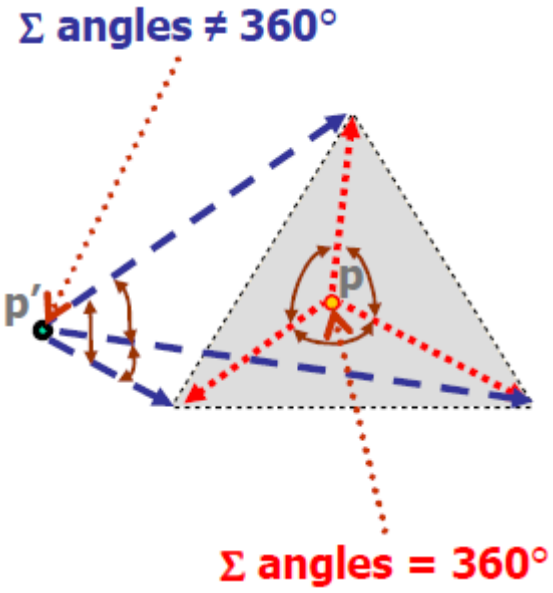
Using sum of angles and dot product

3. If the point is inside the boundaries of the triangle, the sum of the angles must be

$$\theta_1 + \theta_2 + \theta_3 = 360^\circ$$

4. If the point is outside the triangle, then

$$\theta_1 + \theta_2 + \theta_3 \neq 360^\circ$$

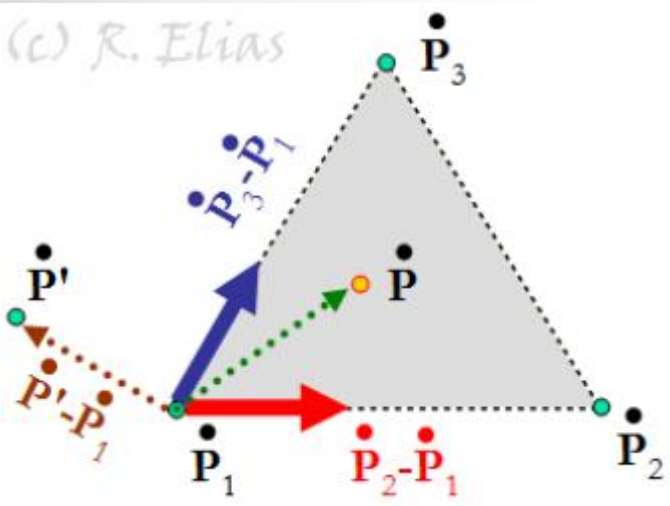


Ray-Triangle Intersection: Point in Triangle II

Using cross and dot products

- Each edge of a triangle splits its plane into two sides.
- Points on one side behave differently than points on the other side.
- To express this mathematically:

$$\mathbf{n}_{p'} = \left[\dot{\mathrm{P}}_3 - \dot{\mathrm{P}}_1\right] \times \left[\dot{\mathrm{P}}' - \dot{\mathrm{P}}_1\right]$$
$$\mathbf{n}_{p} = \left[\dot{\mathrm{P}}_3 - \dot{\mathrm{P}}_1\right] \times \left[\dot{\mathrm{P}} - \dot{\mathrm{P}}_1\right]$$



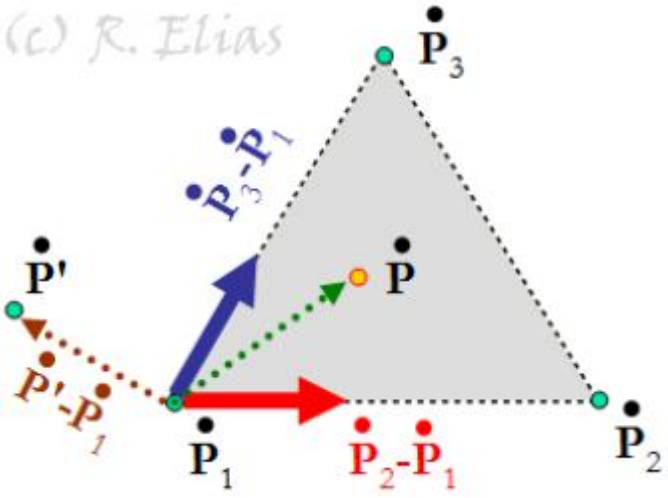
In example shown

- $\dot{\mathrm{P}}$ and $\dot{\mathrm{P}}'$ are on two different sides relative to $\overline{\dot{\mathrm{P}}_1\dot{\mathrm{P}}_3}$.
- $\mathbf{n}_{p'}$ is pointing outwards while $\mathbf{n}_p$ is pointing inwards.
- All points to the left of $\overline{\dot{\mathrm{P}}_1\dot{\mathrm{P}}_3}$ will behave similar to $\dot{\mathrm{P}}'$.
- All points to the right of $\overline{\dot{\mathrm{P}}_1\dot{\mathrm{P}}_3}$ will behave similar to $\dot{\mathrm{P}}$.

Ray-Triangle Intersection: Point in Triangle II

Using cross and dot products

- **Question:** What is the correct direction for the vectors relating to points inside the triangle?
- **Answer:** use the third vertex $\dot{P}_2$ to determine the direction that corresponds to points inside the triangle.



$$\mathbf{n}_{p2} = \left[\dot{P}_3 - \dot{P}_1\right] \times \left[\dot{P}_2 - \dot{P}_1\right]$$

- Now, test the direction of $\mathbf{n}_p$ (or $\mathbf{n}_{p'}$) with respect to $\mathbf{n}_{p2}$. This can be done using the dot product

$$\mathbf{n}_p \bullet \mathbf{n}_{p2} \geq 0 \Rightarrow in$$
$$\mathbf{n}_p \bullet \mathbf{n}_{p2} < 0 \Rightarrow out$$

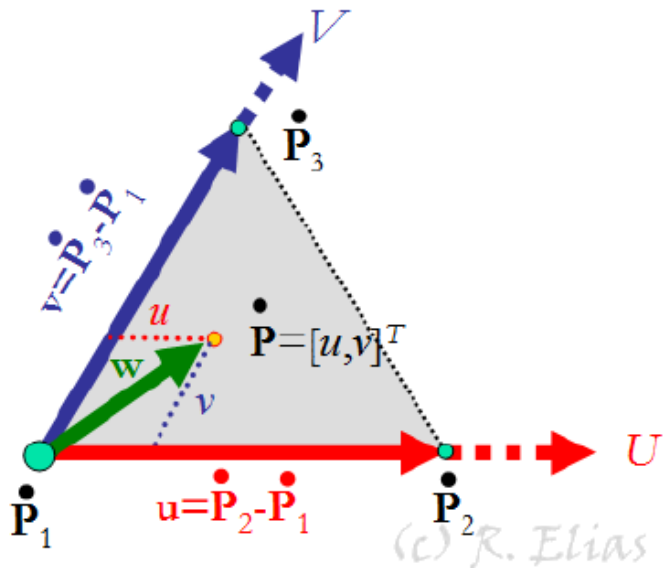
Test **all** sides of the triangle to make sure that the point lies inside.

Ray-Triangle Intersection: Point in Triangle III

Parametric coordinates

1. Pick one of the vertices to be the origin of the plane; e.g., $\dot{\mathrm{P}}_1$.
2. Basis vectors are defined to specify points on the plane (and in the triangle). In the example shown, the basis vectors are

$$\mathbf{u} = \dot{\mathrm{P}}_2 - \dot{\mathrm{P}}_1 \quad \text{and} \quad \mathbf{v} = \dot{\mathrm{P}}_3 - \dot{\mathrm{P}}_1$$



3. A point $\dot{\mathrm{P}}$ can be defined in terms of these basis vectors by starting at the origin $\dot{\mathrm{P}}$ and walking some *u* distance along **u** then walking some *v* distance in the direction **v**. Hence,

$$\dot{\mathrm{P}} = \dot{\mathrm{P}}_1 + u[\dot{\mathrm{P}}_2 - \dot{\mathrm{P}}_1] + v[\dot{\mathrm{P}}_3 - \dot{\mathrm{P}}_1]$$

Ray-Triangle Intersection: Point in Triangle III

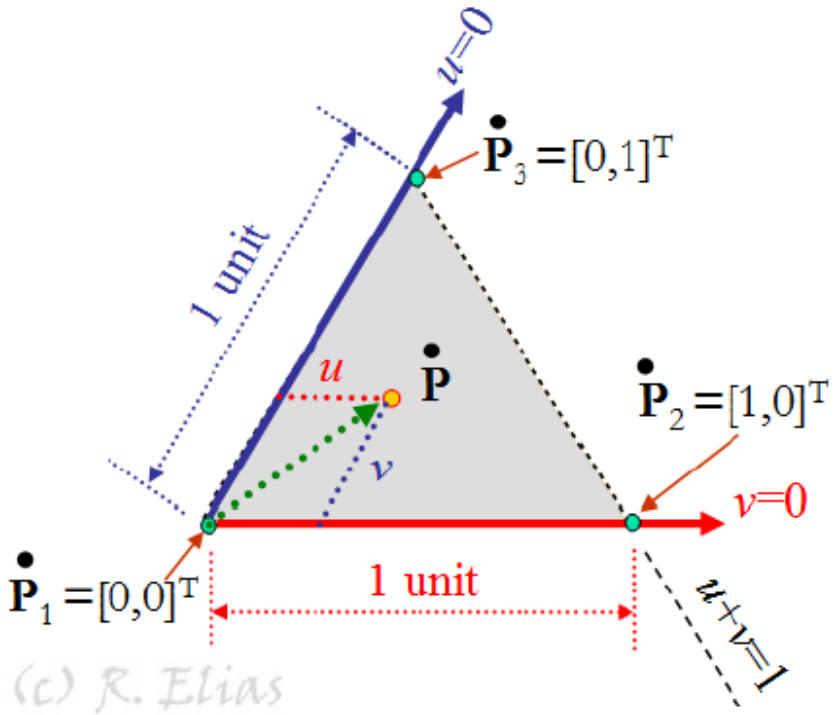
Parametric coordinates

The same is written as

$$[\dot{\mathbf{P}} - \dot{\mathbf{P}}_1] = u[\dot{\mathbf{P}}_2 - \dot{\mathbf{P}}_1] + v[\dot{\mathbf{P}}_3 - \dot{\mathbf{P}}_1]$$
$$\mathbf{w} = u\mathbf{u} + v\mathbf{v}$$

The values of *u* and *v* are

$$u = \frac{(\mathbf{u}\bullet\mathbf{v})(\mathbf{w}\bullet\mathbf{v}) - (\mathbf{v}\bullet\mathbf{v})(\mathbf{w}\bullet\mathbf{u})}{(\mathbf{u}\bullet\mathbf{v})^2 - (\mathbf{u}\bullet\mathbf{u})(\mathbf{v}\bullet\mathbf{v})}$$
$$v = \frac{(\mathbf{u}\bullet\mathbf{v})(\mathbf{w}\bullet\mathbf{u}) - (\mathbf{u}\bullet\mathbf{u})(\mathbf{w}\bullet\mathbf{v})}{(\mathbf{u}\bullet\mathbf{v})^2 - (\mathbf{u}\bullet\mathbf{u})(\mathbf{v}\bullet\mathbf{v})}$$



4. Point $\dot{\mathbf{P}}$ is inside the triangle if **all** the these constraints are true:

$$0 \le u \le 1$$
$$0 \le v \le 1$$
$$u + v \le 1$$

Ray-Triangle Intersection: Point in Triangle IV

Barycentric coordinates

- A point $\dot{\mathrm{P}}$ inside a triangle $\triangle\dot{\mathrm{P}}_1\dot{\mathrm{P}}_2\dot{\mathrm{P}}_3$ splits it into three sub-triangles: $\triangle\dot{\mathrm{P}}\dot{\mathrm{P}}_1\dot{\mathrm{P}}_2$, $\triangle\dot{\mathrm{P}}\dot{\mathrm{P}}_2\dot{\mathrm{P}}_3$ and $\triangle\dot{\mathrm{P}}\dot{\mathrm{P}}_3\dot{\mathrm{P}}_1$.

  whose areas are A_3, A_1 and A_2 respectively.
- The total area of the three sub-triangles (i.e., $A_1 + A_2 + A_3$) is equal to the area of the original triangle (i.e., A).
- Barycentric coordinates of the internal point is given by

$$\frac{A_1}{A} + \frac{A_2}{A} + \frac{A_3}{A} = 1$$

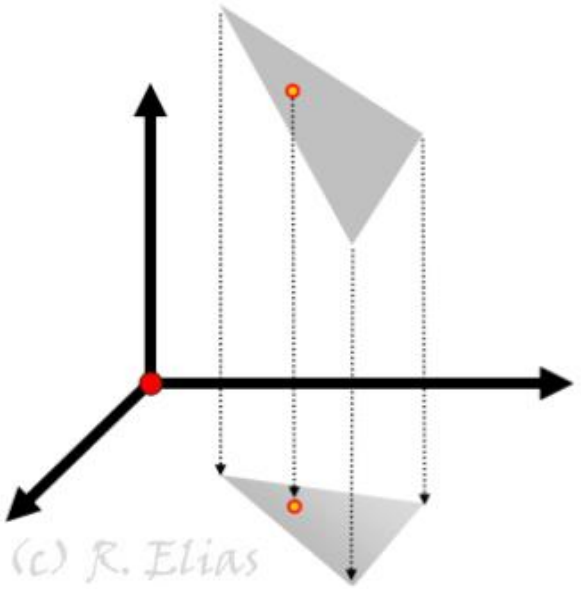
- On the other hand, if the point is outside the triangle

$$\frac{A_1}{A} + \frac{A_2}{A} + \frac{A_3}{A} \neq 1$$

Ray-Triangle Intersection: Point in Triangle V

Using 2D linear equations

- A triangle in 3D space may be projected onto one of the principal planes (i.e., *xy*-, *yz*- or *zx*-plane) along with the tested point in order to eliminate one of the dimensions.



- To achieve this, the coordinate which has the largest component in the plane normal vector may be excluded.
- The problem now is converted to the 2D space. We can use 2D linear equations to test whether a point falls inside the triangle.

Ray-Triangle Intersection: Point in Triangle V

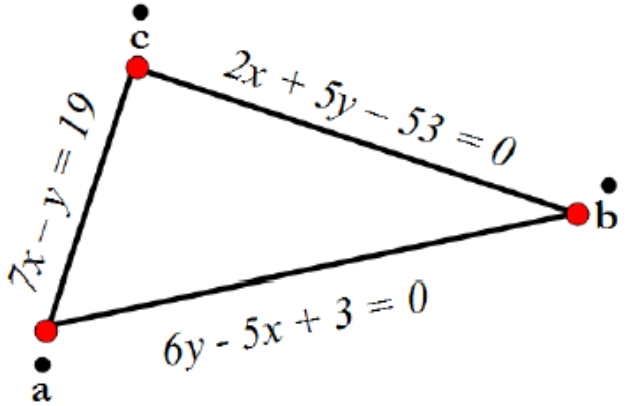
Using 2D linear equations

1. Determine the linear equation for each side of the triangle using its vertices. (Hint: you may use linear equation given two points or cross product of homogeneous points.)
2. For each side, apply the third vertex to its linear equation obtained above and associate the sign of the value calculated to the current side.
3. Apply the point to be tested to the linear equations of the triangle sides.

- A point is inside the triangle if it gives the **same signs** obtained previously when the three vertices are applied to the three linear equations of the sides.

Point in Triangle: An Example

- **Example:** Consider a triangle residing on the *xy*-plane and given as a set of three linear equations. Check if a point $[5, 8]^T$ falls inside this triangle.



- **Answer:**

1. Get the vertices of the triangle as the intersections of its sides.

$$\mathbf{a} = \begin{bmatrix} 7 \\ -1 \\ -19 \end{bmatrix} \times \begin{bmatrix} -5 \\ 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 111 \\ 74 \\ 37 \end{bmatrix} \Rightarrow \dot{\mathbf{a}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

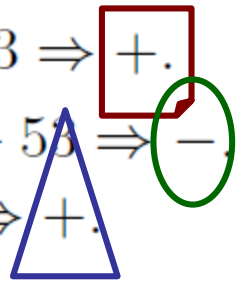
Similarly, $\dot{\mathbf{b}} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$ $\dot{\mathbf{c}} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$

Point in Triangle: An Example

2. Apply vertices to the linear equations.

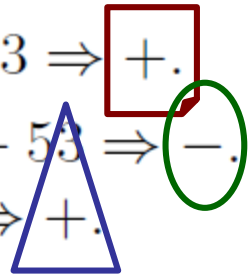
   a) Get the "inside" signs with respect to the edges:

$$\text{Apply } \dot{\mathbf{c}} = [4,9]^T \text{ to } \overline{\dot{\mathbf{a}}\dot{\mathbf{b}}}\text{: } 6y - 5x + 3 \Rightarrow 6 \times 9 - 5 \times 4 + 3 \Rightarrow +.$$
$$\text{Apply } \dot{\mathbf{a}} = [3,2]^T \text{ to } \overline{\dot{\mathbf{b}}\dot{\mathbf{c}}}\text{: } 2x + 5y - 53 \Rightarrow 2 \times 3 + 5 \times 2 - 53 \Rightarrow -.$$
$$\text{Apply } \dot{\mathbf{b}} = [9,7]^T \text{ to } \overline{\dot{\mathbf{c}}\dot{\mathbf{a}}}\text{: } 7x - y - 19 \Rightarrow 7 \times 9 - 7 - 19 \Rightarrow +.$$



   b) Check if the point [5, 8]^T falls inside this triangle:

$$\text{Apply } [5,8]^T \text{ to } \overline{\dot{\mathbf{a}}\dot{\mathbf{b}}}\text{: } 6y - 5x + 3 \Rightarrow 6 \times 8 - 5 \times 5 + 3 \Rightarrow +.$$
$$\text{Apply } [5,8]^T \text{ to } \overline{\dot{\mathbf{b}}\dot{\mathbf{c}}}\text{: } 2x + 5y - 53 \Rightarrow 2 \times 5 + 5 \times 8 - 53 \Rightarrow -.$$
$$\text{Apply } [5,8]^T \text{ to } \overline{\dot{\mathbf{c}}\dot{\mathbf{a}}}\text{: } 7x - y - 19 \Rightarrow 7 \times 5 - 8 - 19 \Rightarrow +.$$



Same signs ➔ The point [5, 8]^T falls **inside** this triangle.

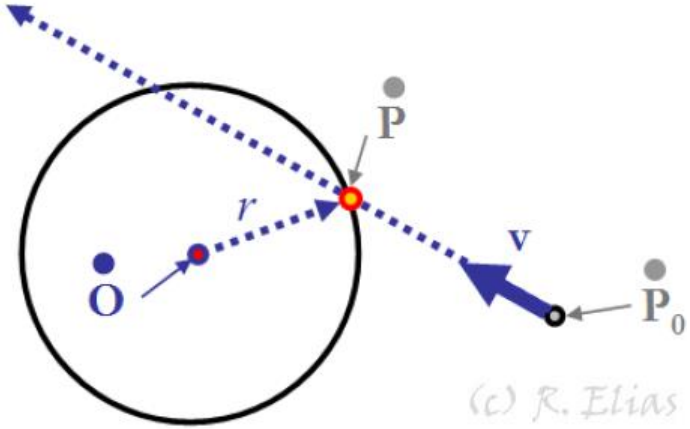
Ray-Sphere Intersection: Algebraic Method

Zero to two possible solutions!

Ray equation $\dot{\mathbf{P}} = \dot{\mathbf{P}}_0 + t\mathbf{v}$

Sphere equation $\left\|\dot{\mathbf{P}} - \dot{\mathbf{O}}\right\|^2 = r^2$

Substituting $\left\|\left[\dot{\mathbf{P}}_0 + t\mathbf{v}\right] - \dot{\mathbf{O}}\right\|^2 = r^2$



=

$$\underbrace{\|\mathbf{v}\|^2}_{a} t^2 + \underbrace{\left(2\mathbf{v} \bullet \left[\dot{\mathbf{P}}_0 - \dot{\mathbf{O}}\right]\right)}_{b} t + \underbrace{\left\|\dot{\mathbf{P}}_0 - \dot{\mathbf{O}}\right\|^2 - r^2}_{c} = 0$$

Once *t* is obtained, substitute in ray equation.

Ray-Sphere Intersection: Geometric Method

Zero to two possible solutions!

$$\mathbf{a} = \dot{\mathbf{O}} - \dot{\mathbf{P}}_0$$

$$\delta_1 = \mathbf{a} \bullet \mathbf{v}$$

$$\delta_3 = \sqrt{\mathbf{a} \bullet \mathbf{a} - \delta_1^2}$$

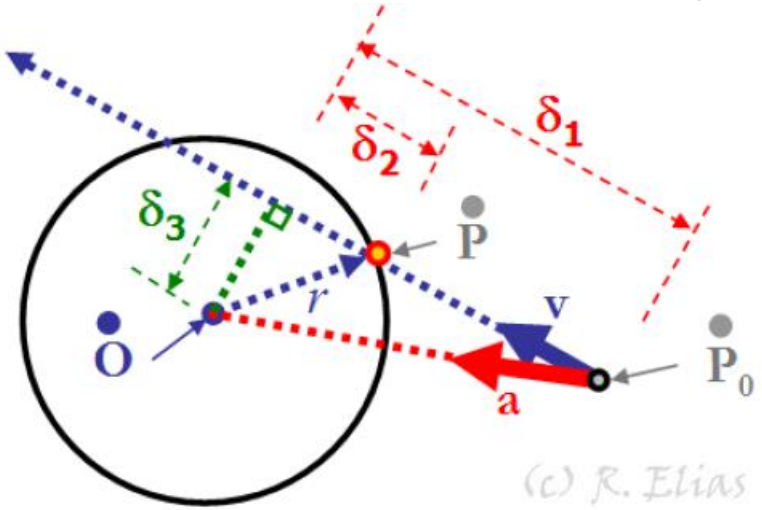
$$\delta_2 = \sqrt{r^2 - \delta_3^2}$$

$$t = \delta_1 - \delta_2$$

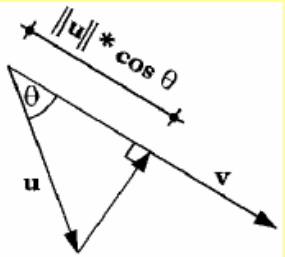
$$t = \mathbf{a} \bullet \mathbf{v} - \sqrt{r^2 - \left(\mathbf{a} \bullet \mathbf{a} - (\mathbf{a} \bullet \mathbf{v})^2\right)}$$

$$\dot{\mathbf{P}} = \dot{\mathbf{P}}_0 + t\mathbf{v}$$

$$\dot{\mathbf{P}} = \dot{\mathbf{P}}_0 + \left(\mathbf{a} \bullet \mathbf{v} - \sqrt{r^2 - \left(\mathbf{a} \bullet \mathbf{a} - (\mathbf{a} \bullet \mathbf{v})^2\right)}\right)\mathbf{v}$$



Aside



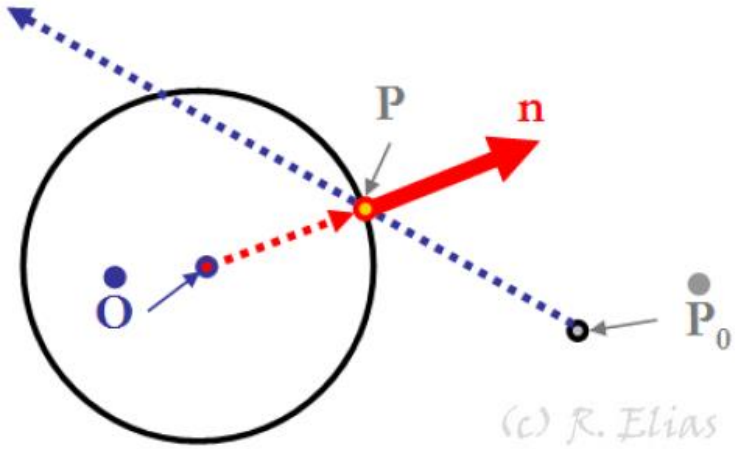
The length of the projection of **u** onto **v** is

$$\frac{\mathbf{u} \bullet \mathbf{v}}{\|\mathbf{v}\|}$$

Ray-Sphere Intersection: Normal Vector

- The normal vector at the point of intersection between a ray and a sphere can be calculated as

$$\mathbf{n} = \dot{\mathrm{P}} - \dot{\mathrm{O}}$$

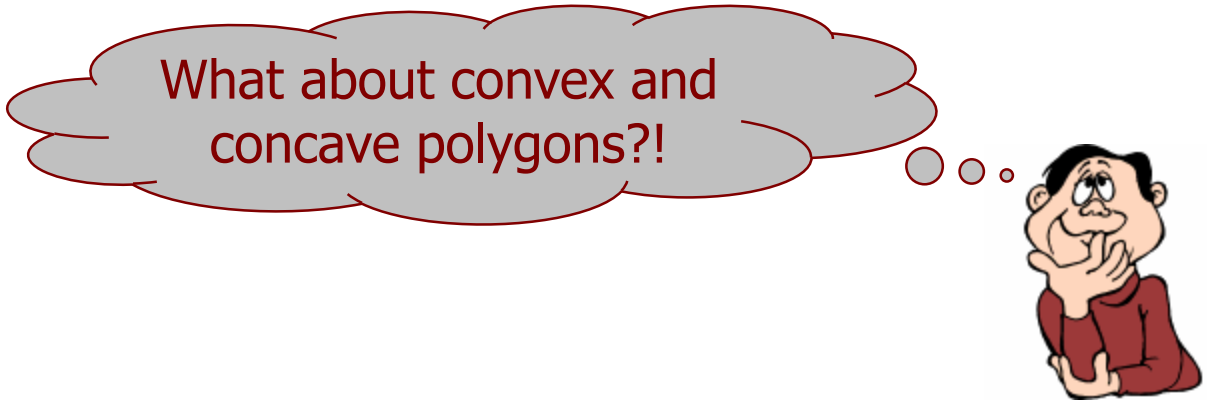


- Then the normalized vector is expressed as

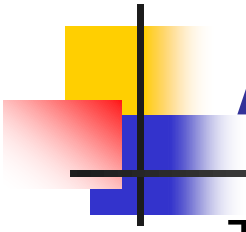
$$\hat{\mathbf{n}} = \frac{\dot{\mathrm{P}} - \dot{\mathrm{O}}}{\|\dot{\mathrm{P}} - \dot{\mathrm{O}}\|}$$

Ray Intersections with Other 3D Objects

- Other 3D objects may include
  - Cones, cylinders: Think about how we tested the ray with the sphere!
  - Boxes: Apply the idea of ray-plane intersection to the sides and return the closest.



Accelerating the Process

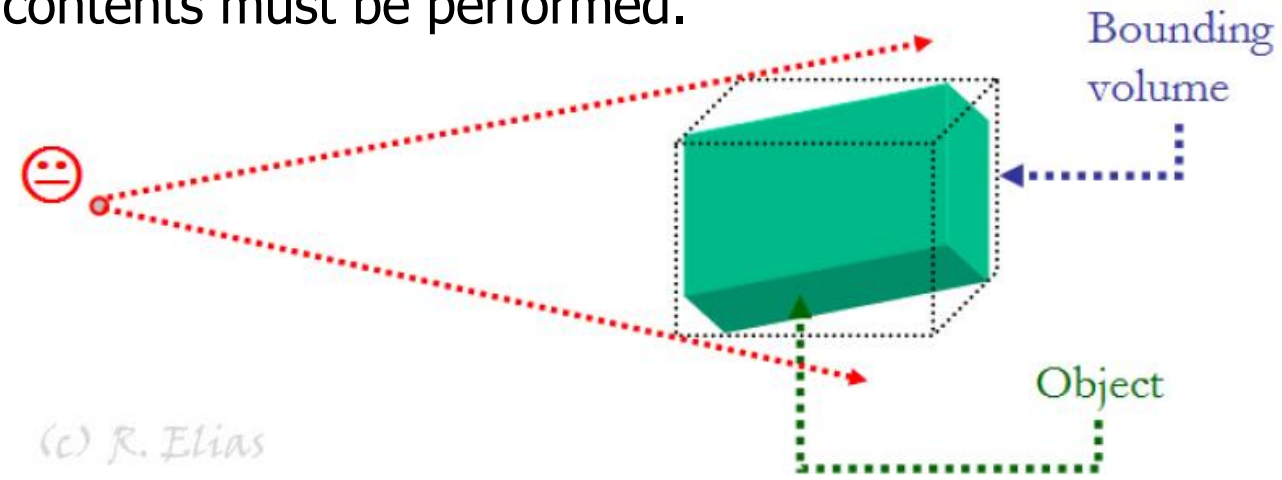


- Testing for intersections should not be performed for all objects.
- Many objects may not exist in the path of a ray and should not be tested for intersections.
  - Such trivial rejections make the process faster.

- We may use
  - bounding volume technique and
  - Spatial partitioning techniques:
    - Regular grid of equal-sized units.
    - Quadtrees.
    - Octrees.
  - …

Bounding Volume

- In this technique, the ray is tested for intersections with the bounding volumes of objects.
  - If the ray **does not hit** the bounding volume, the contained object should be excluded from further processing (as the ray will not hit its contents).
  - If the ray **hits** the bounding volume, further testing for the contents must be performed.

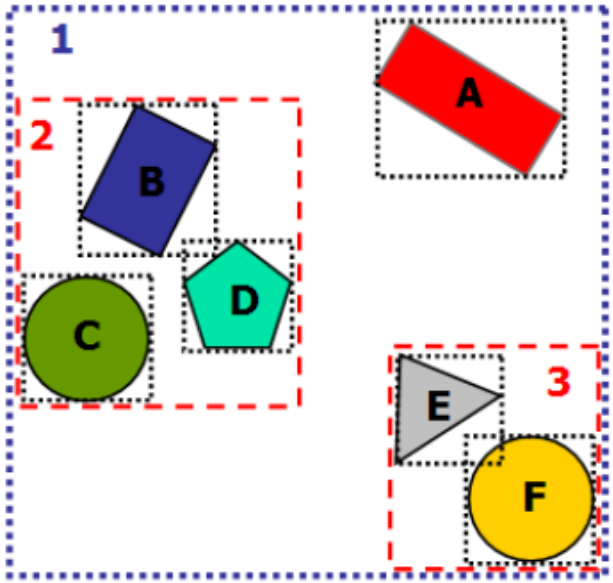


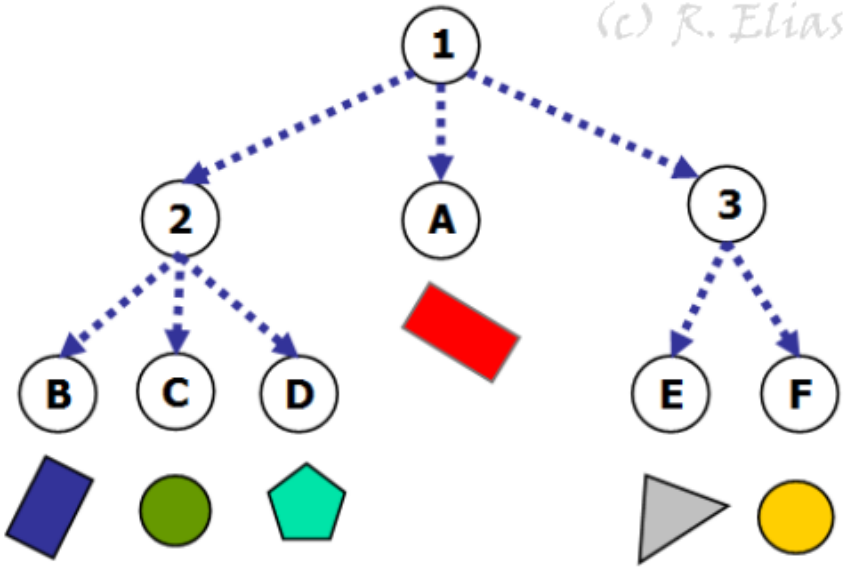
Bounding Volume Hierarchies

- Bounding volumes may overlap.
- Overlapping bounding volumes may be grouped into a larger bounding volume.
- This can be set as a hierarchy where a larger bounding volume acts as a parent and the contained smaller bounding volumes along with their objects become its children.

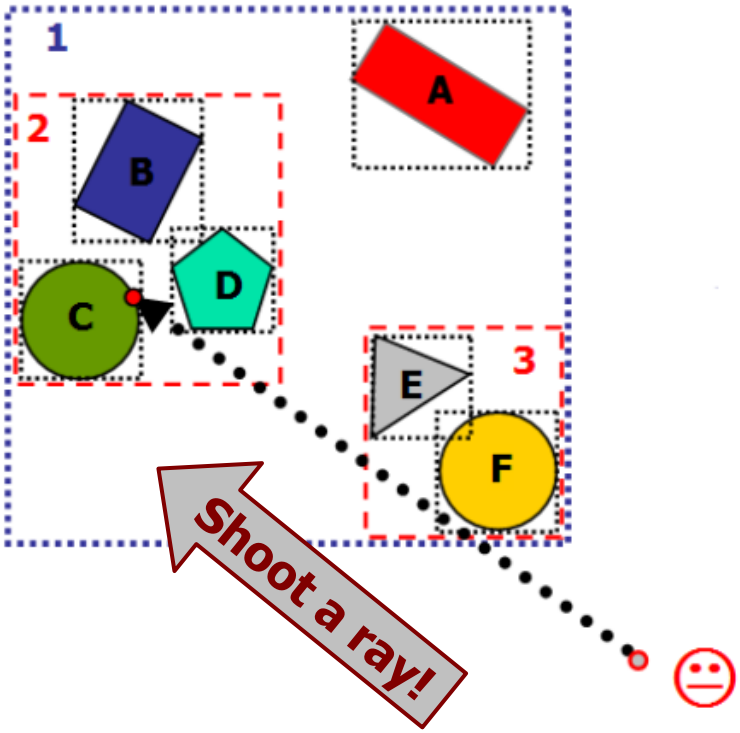
- **How does this affect the acceleration of the process?**
  - If the ray **does not hit** the bounding volume of the parent, all children will be **excluded** from further processing (because children will not be along the path of the ray).

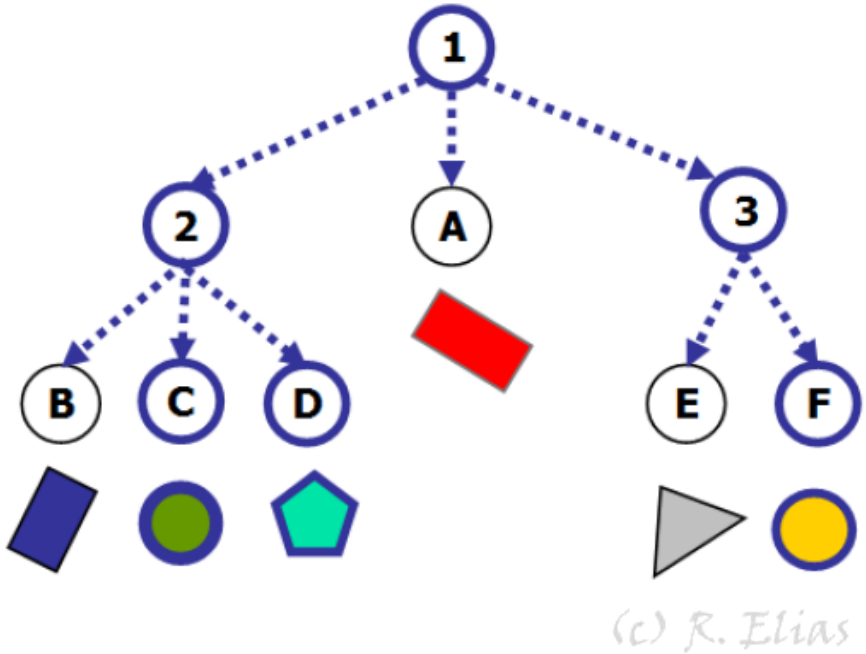
Bounding Volume Hierarchies: A 2D Example



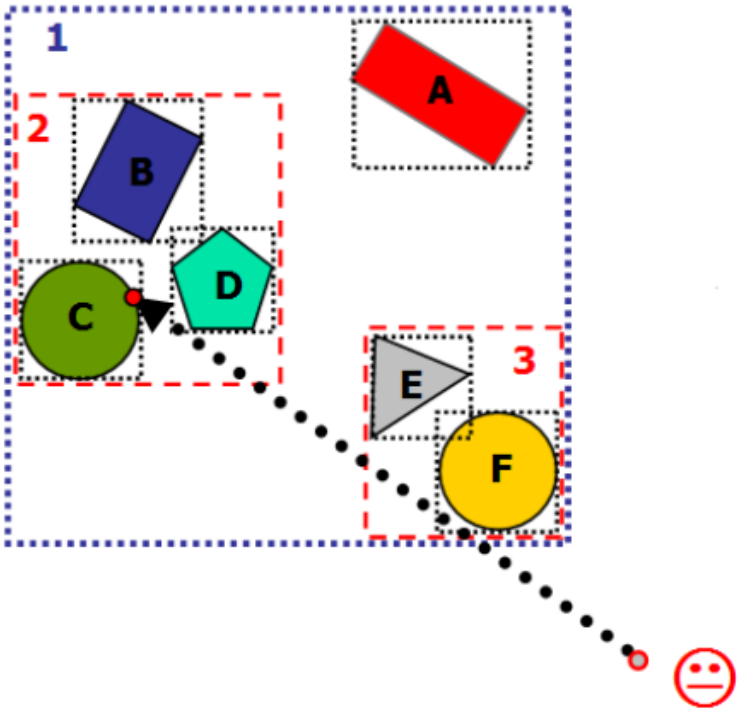


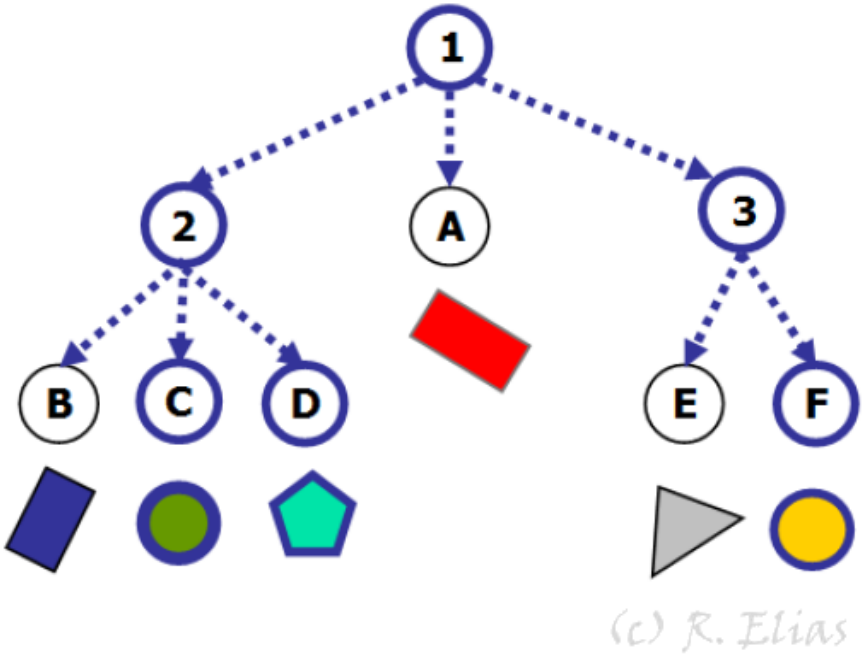
Bounding Volume Hierarchies: A 2D Example





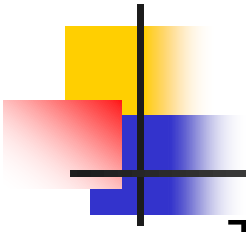
Bounding Volume Hierarchies: A 2D Example





- Steps! 1 3 F 2 D C

Space Partitioning: Regular Grid



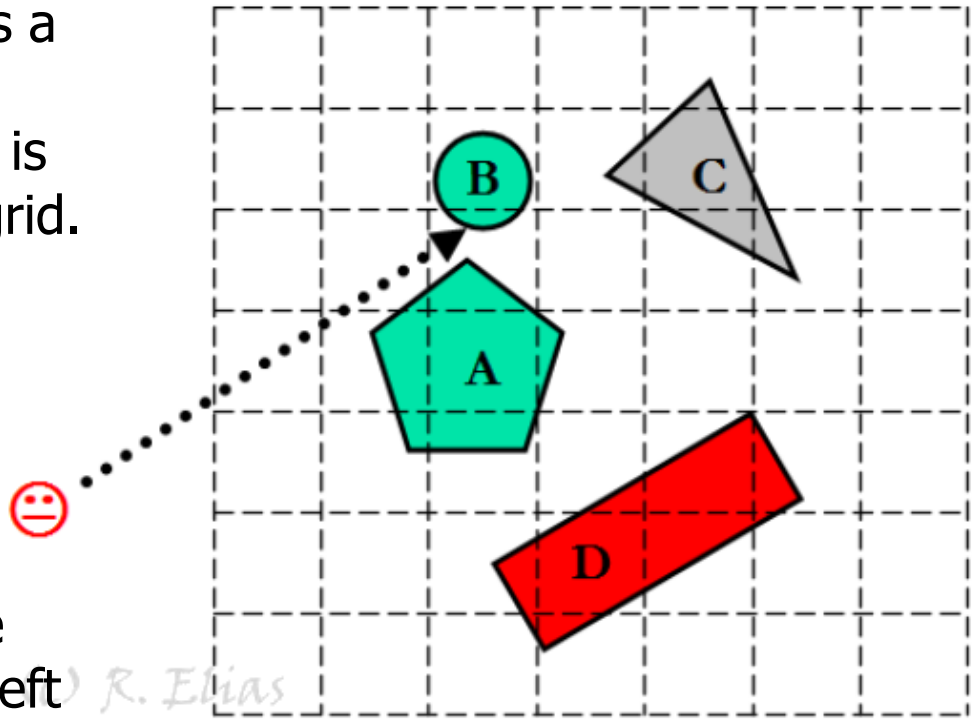
- The idea is to partition the bounding volume of the whole space using a regular grid of equal-sized units in a top-down process.
- A list of objects is then prepared for each partition.
  - A partition list may contain zero, one or more objects while an object may be extended along one or more partitions.
- When a ray is shot through the space, it needs to be tested for intersections with objects residing in the partitions it passes through.
- Examining partitions should be done in the order in which the ray travels.

Space Partitioning: Regular Grid

- Once an intersection is found, no more partitions along the ray are tested.
  - Notice, however, that in some situations, a partition may contain more than one object and more than one intersection may be found. Hence, all intersections must be detected while the closest one is selected.

Space Partitioning: Regular Grid

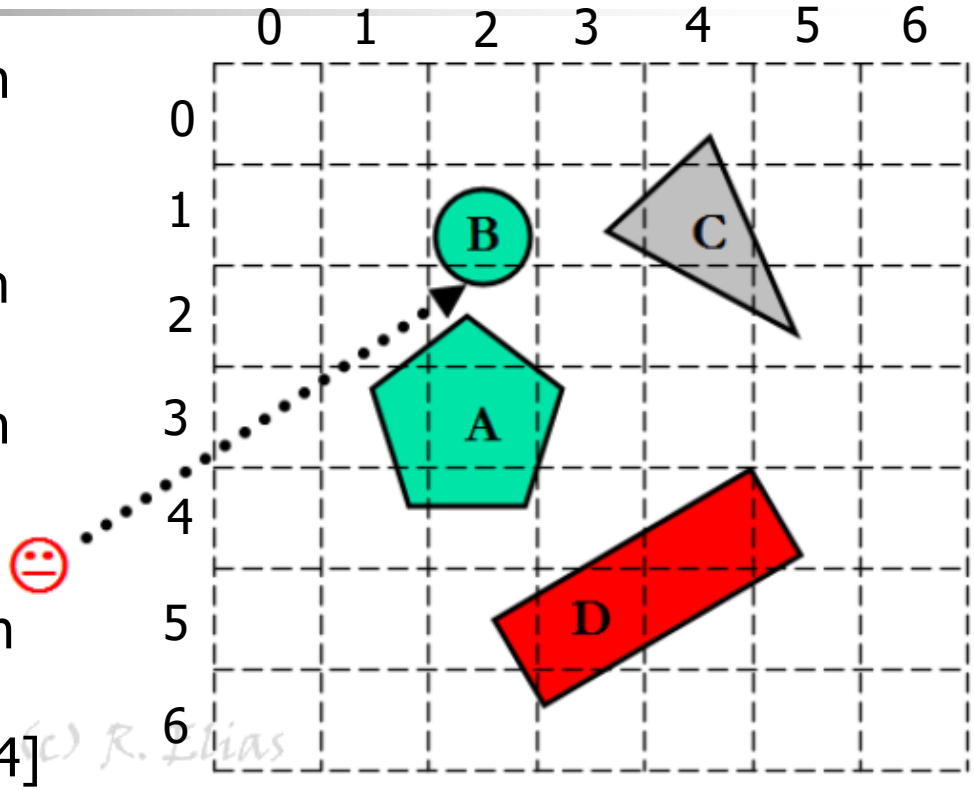
- **Example:** This figure shows a static 2D space containing some 2D shapes. The space is partitioned using a regular grid. If a ray is shot as indicated, determine the sequence of partitions tested for intersections.



- **Answer:** If the origin of the regular grid is at the upper left corner, the objects and associated partitions will be:

Space Partitioning: Regular Grid

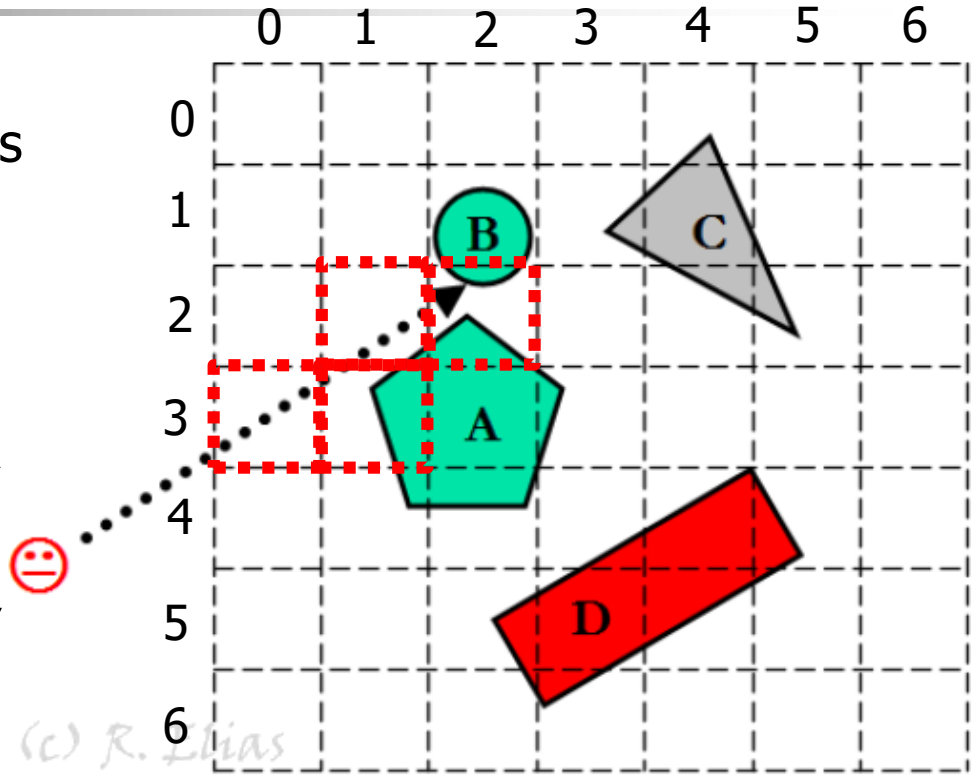
- Object "A" is associated with partitions [1,2], [1,3], [1,4], [2,2], [2,3], [2,4] and [3,3].
- Object "B" is associated with partitions [2,1] and [2,2].
- Object "C" is associated with partitions [3,1], [4,0], [4,1], [4,2], [5,1] and [5,2].
- Object "D" is associated with partitions [2,5], [2,6], [3,4], [3,5], [3,6], [4,4], [4,5], [5,4] and [5,5].
- All other partitions are empty.



Done only **once** for a static scene!

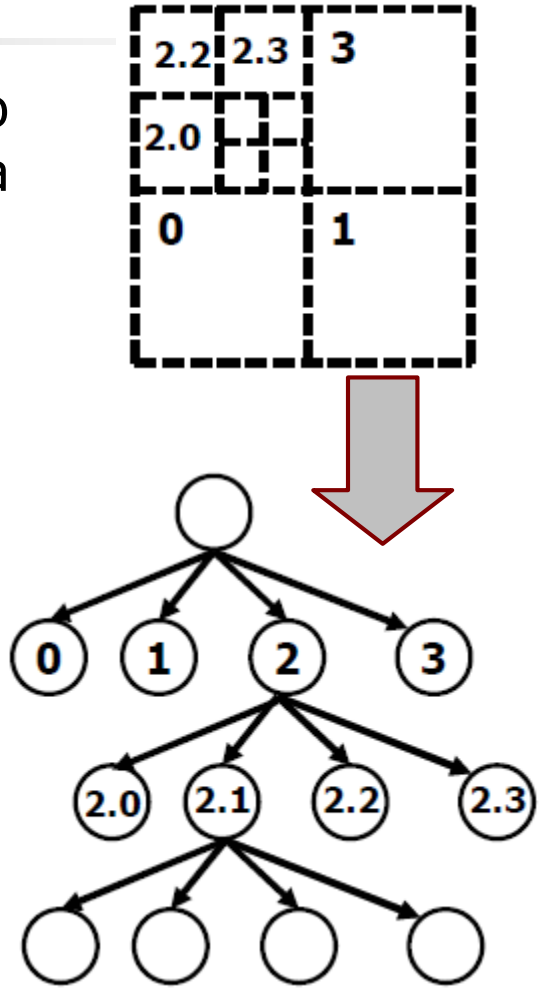
Space Partitioning: Regular Grid

- When the ray is shot, it passes through the partitions [0,3], [1,3], [1,2] and then [2,2].
  - **[0,3]:** Empty ➔ no intersection
  - **[1,3], [1,2]:** Object "A" ➔ no intersection
  - **[2,2]:** Objects "A" & "B"
    - Object "A" ➔ no intersection
    - Object "B" ➔ intersection detected



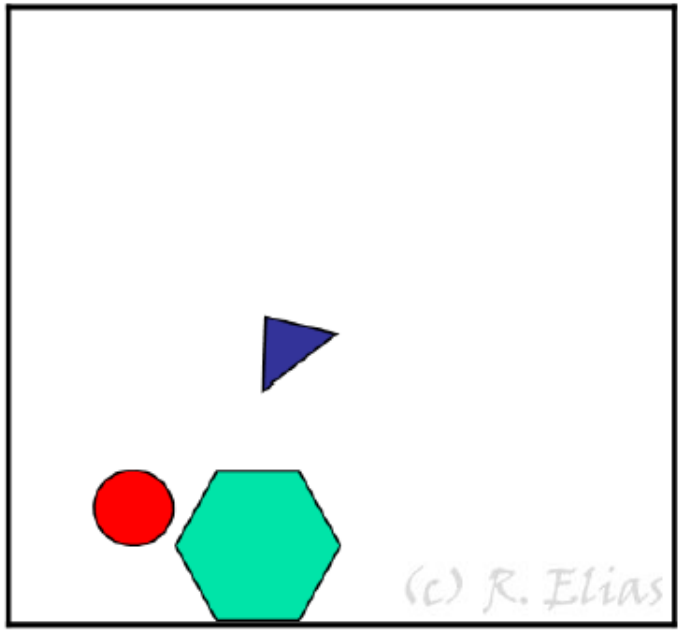
Space Partitioning: Quadtrees

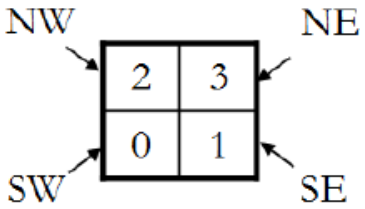
- A **quadtree** is used to partition a 2D space into four quadrants, which represent 4 children for a root node that represents the whole space.
- If a child is not empty or full, it may be split again into 4 quadrants representing four children of that node.
- The splitting continues until we reach a stopping condition; e.g.,
  - the leaf node becomes either empty or full
  - a number of subdivisions is reached
  - a number of objects in a node is obtained
  - …



Space Partitioning: Quadtrees

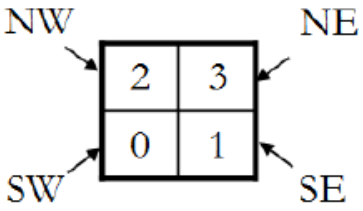
- **Example:** This figure shows a 2D space containing some 2D shapes. It is required to partition this space using a quadtree such that a shape cannot be split between partitions and a partition cannot contain more than one shape. The order of quadrants is

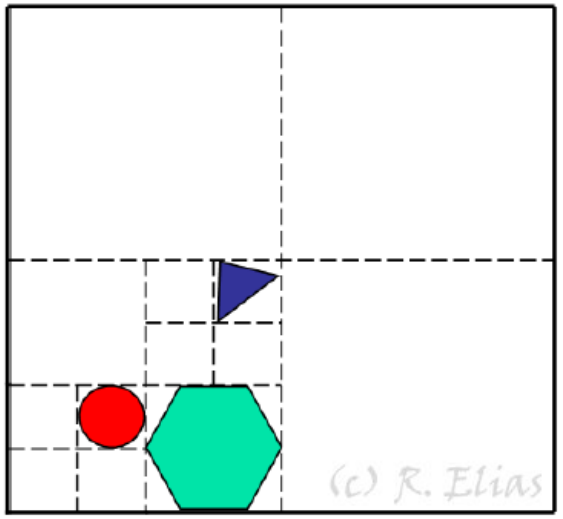


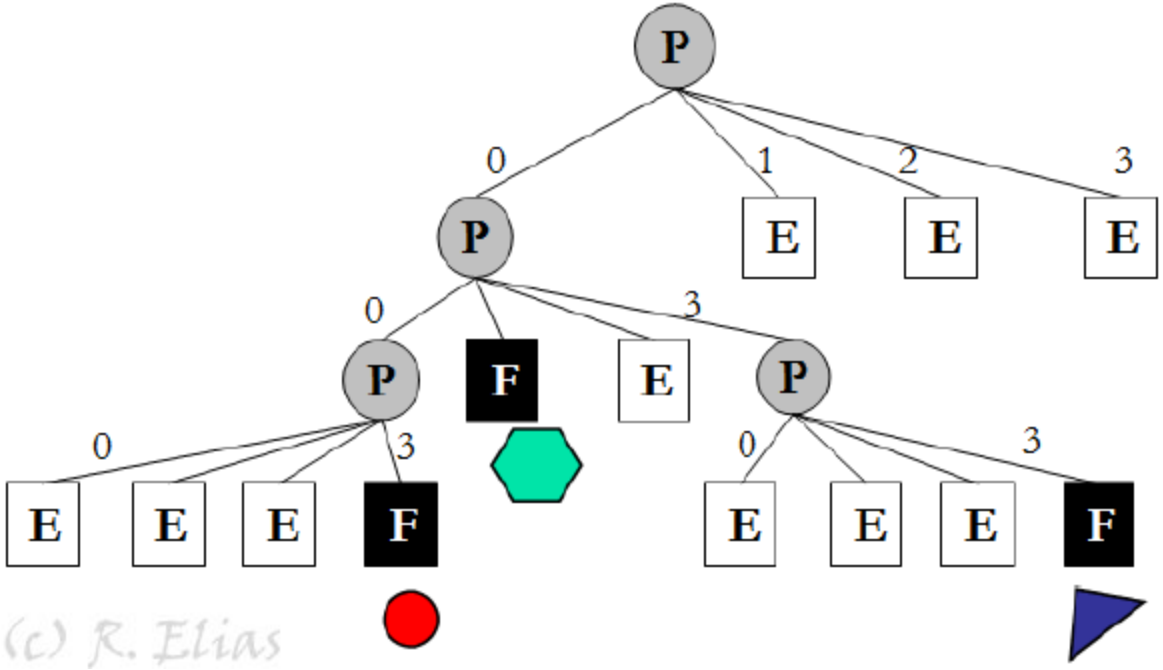


Space Partitioning: Quadtrees

- **Answer:**

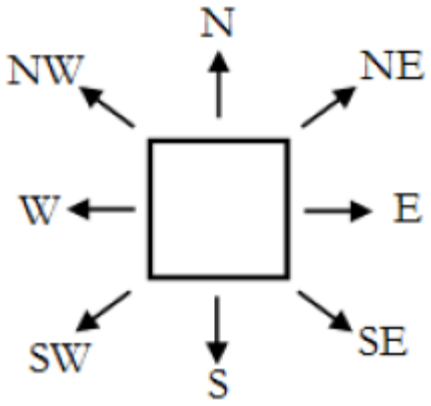






Space Partitioning: Quadtree Neighboring Nodes

- **Given a ray shot through a direction into a space represented by a quadtree, how can we use such a tree structure to specify the partitions the ray passes through?**
- There are only 8 different directions a ray may follow from a certain partition.
- Going from node "A" to node "B", this neighbor finding problem is solved as:



1. Start at node "A" and go up until a common ancestor with node "B" is found.
2. From the common ancestor, go down the quadtree until node "B" is found. This descending process follows a mirror path from node "A" to the ancestor that is reflected about the common border indicated by the given direction.

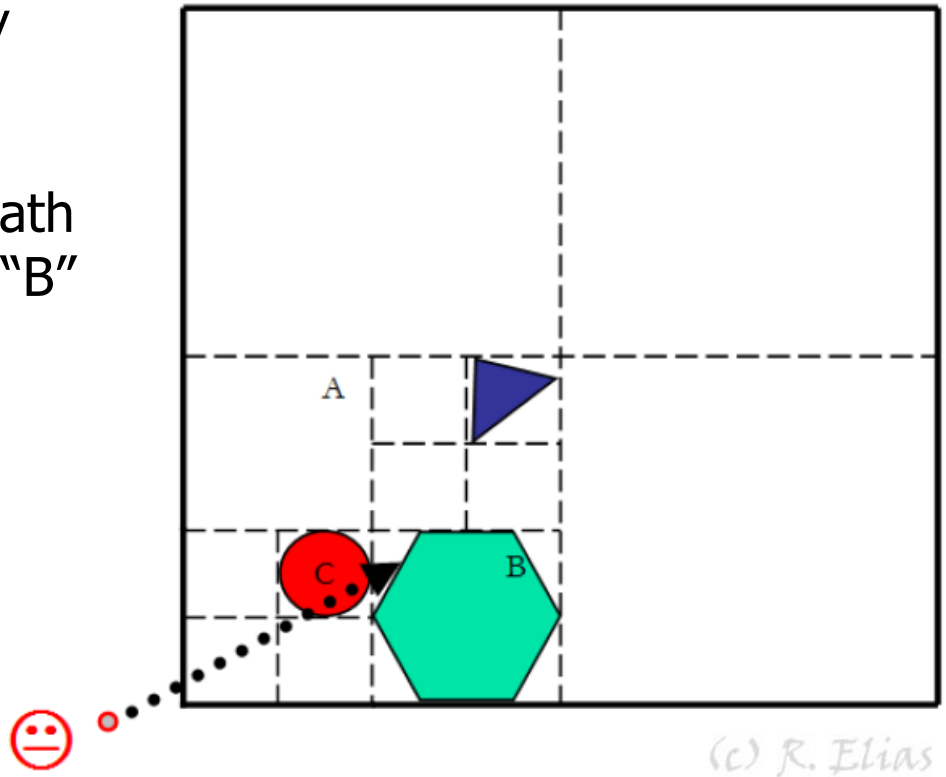
Space Partitioning: Quadtree Neighboring Nodes

Note that

- The **common ancestor** is the first node encountered that is not reached from a child on the given direction side of the node.
- The **common border**, for example, is N-S for directions N and S and it is SW-NE for directions SW and NE.

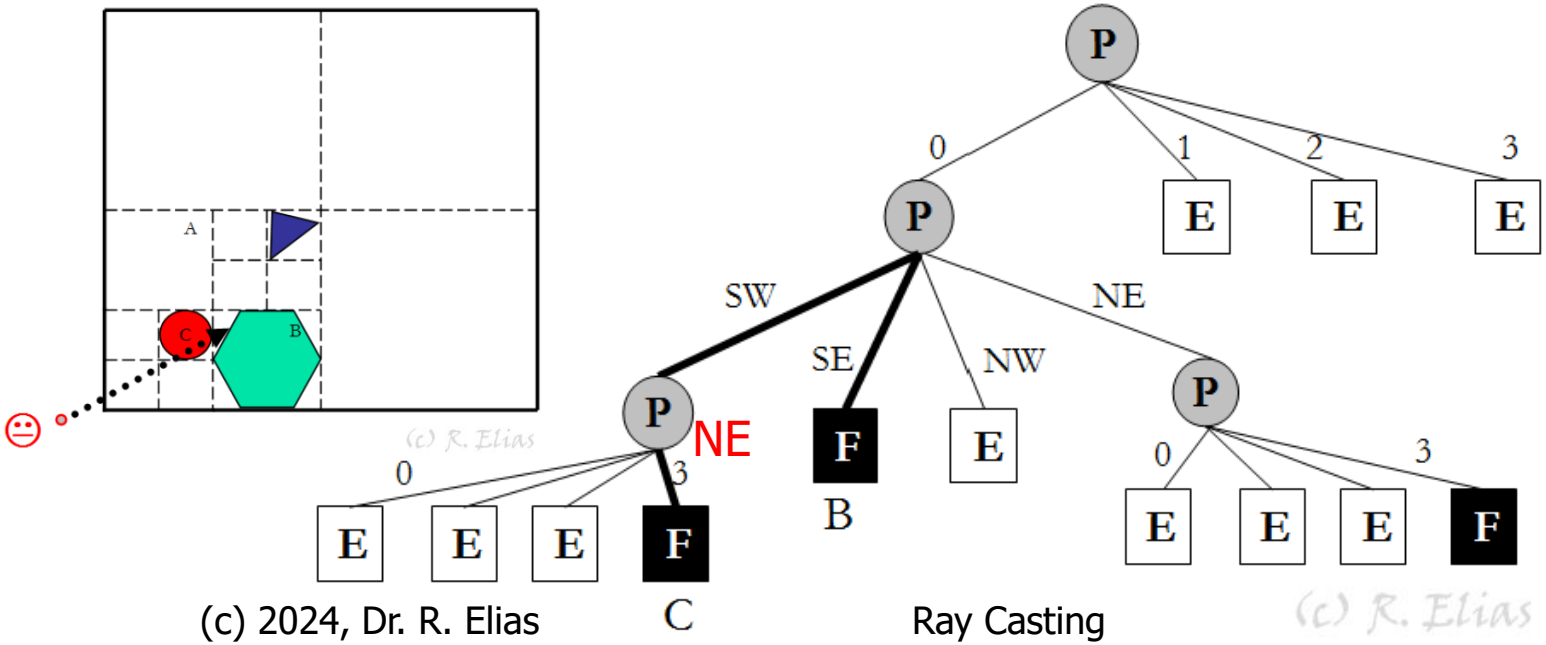
Space Partitioning: Quadtree Neighboring Nodes

- **Example:** In this space, a ray passes from partition "C" to partition "B" through their shared edge. Determine the path from partition "C" to partition "B" on the quadtree using the neighbor finding method.



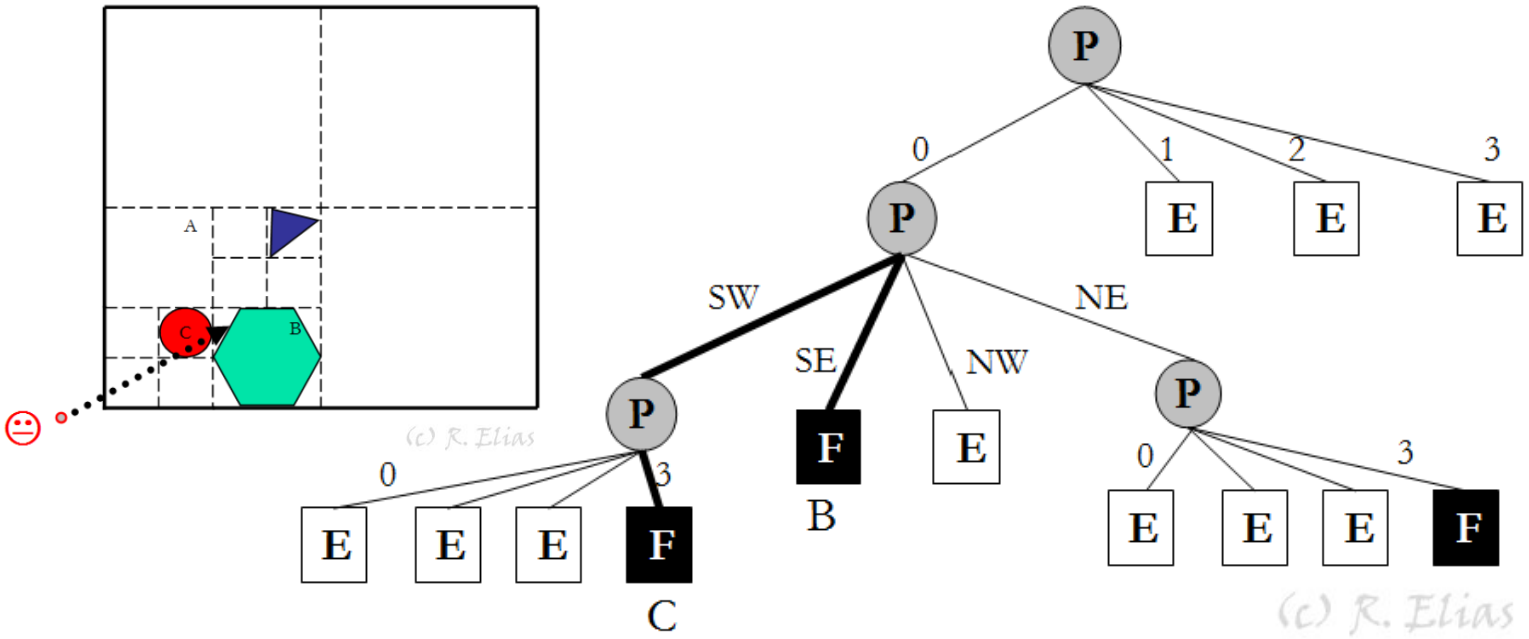
Space Partitioning: Quadtree Neighboring Nodes

- **Answer:** Go up from partition "C" to its parent. Thus, we reach this ancestor from an NE child (i.e., an E node) while the direction is E.
- Hence, go one step upwards to its grandparent. We reach this ancestor from an SW child (i.e., a W node) while the direction is E. Thus, we stop at this parent.



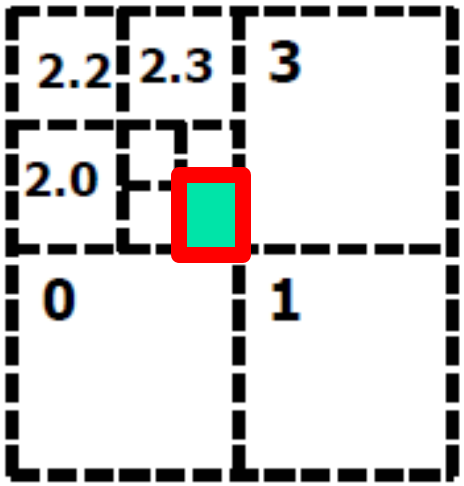
Space Partitioning: Quadtree Neighboring Nodes

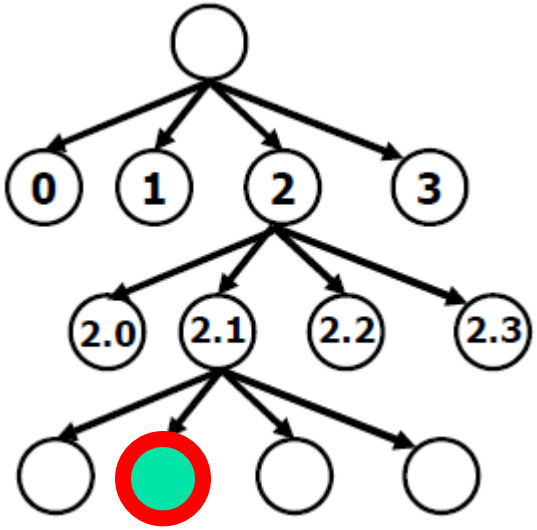
- Go down the quadtree by reflecting about the W-E border; thus, we reflect from SW to SE to reach a leaf node representing partition "B."



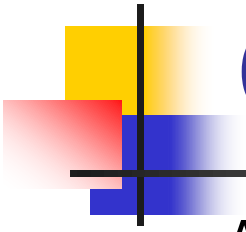
Space Partitioning: Quadtree Neighboring Nodes

- **Example:** If a ray direction is S, determine the neighbor of 2.1.1.
- **Answer:** Left as an exercise!





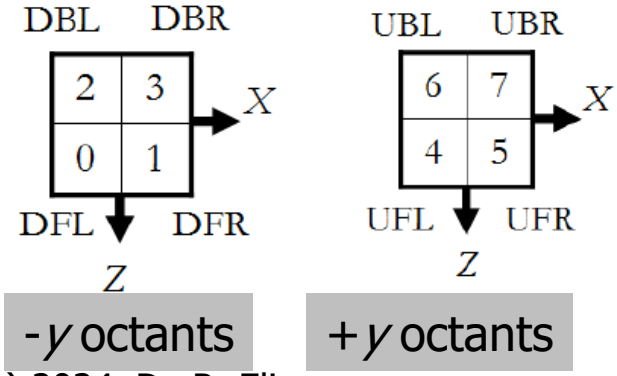
Space Partitioning: Octrees



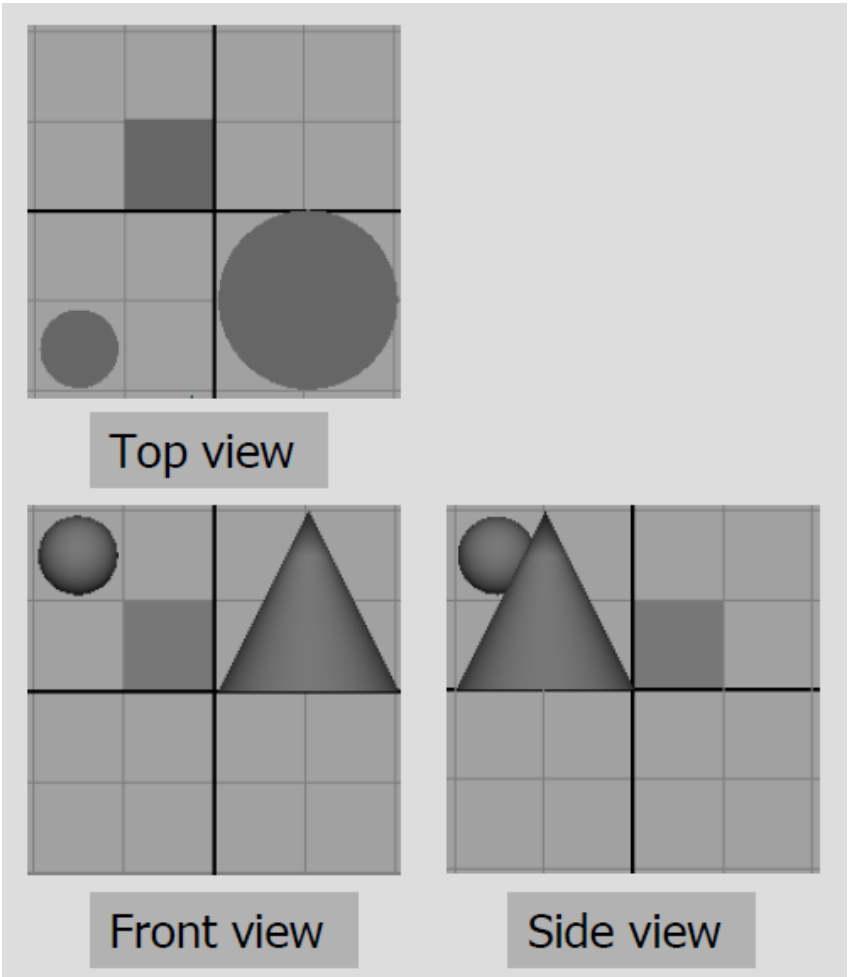
- An **octree** (oct + tree) is a tree data structure where any node (except for the leaves) has 8 children.
- The partially full 3D space is split into 8 octants using 3 planes such that each of the 8 octants is assigned to one of the children.
- The splitting continues recursively as long as a node is partially full.
- This recursive process halts when a stopping condition is met (e.g., all leaf nodes are either full or empty or the maximum number of levels has been reached, etc.).

Space Partitioning: Octrees

- **Example:** A 3D scene is shown as multi-view projections. If this space is partitioned through 3 perpendicular planes using an octree such that each leaf node is a cube that is either empty or containing a whole object. Construct the octree if the order of constructing nodes is:

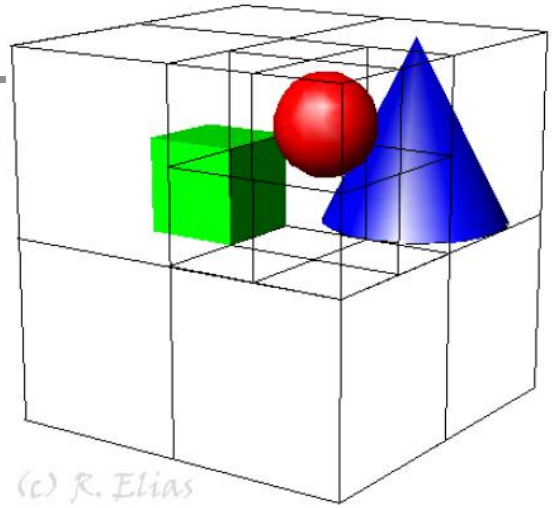


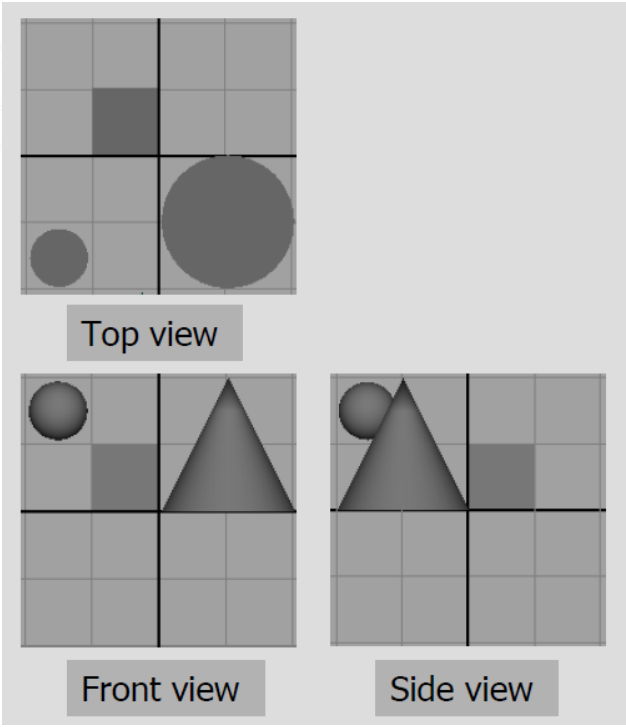
-y octants +y octants

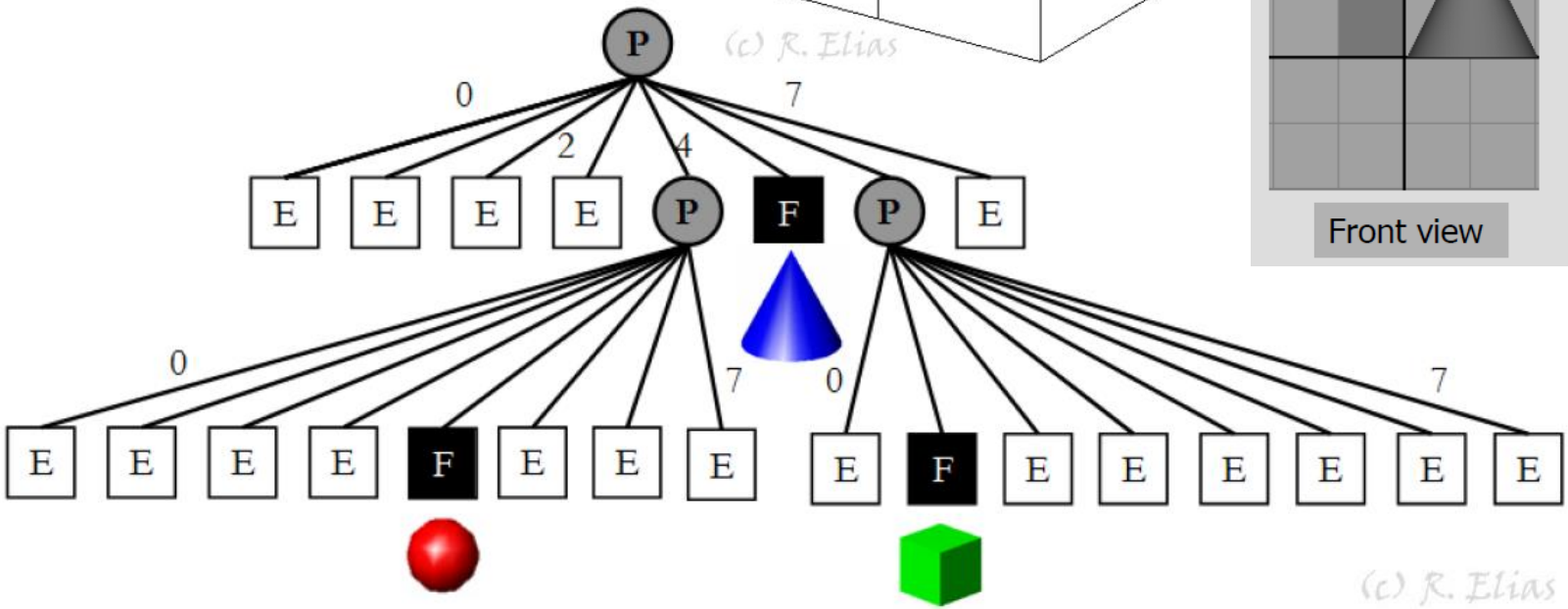


Space Partitioning: Octrees

- **Answer:**



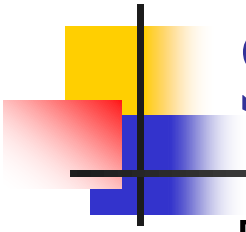




(c) R. Elias

Space Partitioning: Octree Neighboring Nodes

- When a ray is shot, we can specify the partitions the ray passes through using such a tree.
- There are 26 possible directions a ray may follow from a certain partition:
  - 6 neighbors share a face
  - 12 neighbors share an edge
  - 8 neighbors share a vertex
- Finding an octree path from one partition to another is similar to the approach discussed previously.



Summary

- Ray casting
  - Rays through pixels
  - Ray intersections
    - Ray-plane intersection
    - Ray-sphere intersection
    - Ray-triangle intersection
  - Accelerating the process
    - Bounding volume
    - Binary space partitioning
    - Quadtrees
    - Octrees